nature communications

Article

https://doi.org/10.1038/s41467-024-52209-5

Stereoselective rhodium-catalyzed reaction of allenes with organoboronic reagents for diversified branched 1,3-alkadienes

Received: 19 June 2024

Accepted: 28 August 2024

Published online: 27 September 2024

Check for updates

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The terminal isoprene unit, as the simplest branched 1,3-diene unit, exists in a wide range of natural products and bioactive molecules. Herein, we report a stereoselective rhodium-catalyzed reaction of allenes with readily available methyl pinacol boronic ester, providing a straightforward approach to isoprene derivatives with a very high *E*-stereoselectivity. Its synthetic potential has been illustrated by a concise synthesis of natural product schinitrienin. Such a protocol can be easily extended to aryl and alkenyl boronic reagents affording 2-aryl or -alkenyl substituted 1,3-dienes, which are also of high importance in organic synthesis but remain challenging for their selective synthesis, with a remarkable stereoselectivity. A series of deuterium-labeling experiments indicate a unique mechanism, which involves reversible β-H elimination as well as hydrometalation and isomerization of the allylic rhodium species.

The terminal isoprene unit, as an iconic type of 1,3-dienes, exists in a wide range of natural products and bioactive molecules (Fig. 1a)[1–6]. Featured by the methyl group on the 2-position, the terminal isoprene unit is the simplest branched 1,3-diene unit. However, although the synthesis of linear 1,3-dienes has been well-developed, the stereoselective synthetic methods for branched 1,3-dienes are still very limited[7–9]. Allenes have been developed as versatile synthetic building blocks for the synthesis of various types of olefins due to the unique and diverse reactivity of their two cumulated carbon-carbon double bonds[10–18]. Branched 1,3-dienes can be synthesized from α-functionalized allenes and aryl/alkenyl metallic species, which requires preinstallation of a leaving group in allene (Fig. 1b)[19–32]. For common allenes without a leaving group, the carbometalation of allenes with organometallic species would generate alkenyl metallic species or allylic metallic species[14], and the β-H elimination of allylic metallic species would, in principle, afford the targeted branched 1,3-dienes (Fig. 1c). Previously reported oxidative addition of aryl/alkenyl halides with Pd(0)[33–38] and aryl/alkenyl C–H activation applying the directing group[39,40], are all not applicable for the methylation to form the terminal isoprene unit existing in numerous natural products as shown in Fig. 1a. We envisioned that a strategy of transmetalation of readily commercially available organoboronic reagents with a proper metal catalyst may address the issues for the efficient synthesis of branched 1,3-dienes from allenes: (1) achieving the desired methylation for natural product syntheses; (2) mild conditions, thus, resulting in a very promising selectivity. Herein, we report a Rh-catalyzed reaction of allenes with organoboronic reagents, using a copper salt as the cocatalyst with air as the green oxidant to complete the catalytic cycle, affording branched 1,3-dienes with a high stereoselectivity and a broad substrate scope under very mild conditions (Fig. 1d). Methylation has been realized and successfully applied to the concise synthesis of natural product schinitrienin.

Results and discussion

We initially focused on the construction of the iconic terminal isoprene unit existing in numerous natural products from allenes, therefore, the reactions of allene **1a** with methyl pinacol boronic ester **2a** were evaluated (Table 1). It was found that in the presence of $[Cp^*RhCl_2]_2$

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https://doi.org/10.1038/s41467-024-52209-5

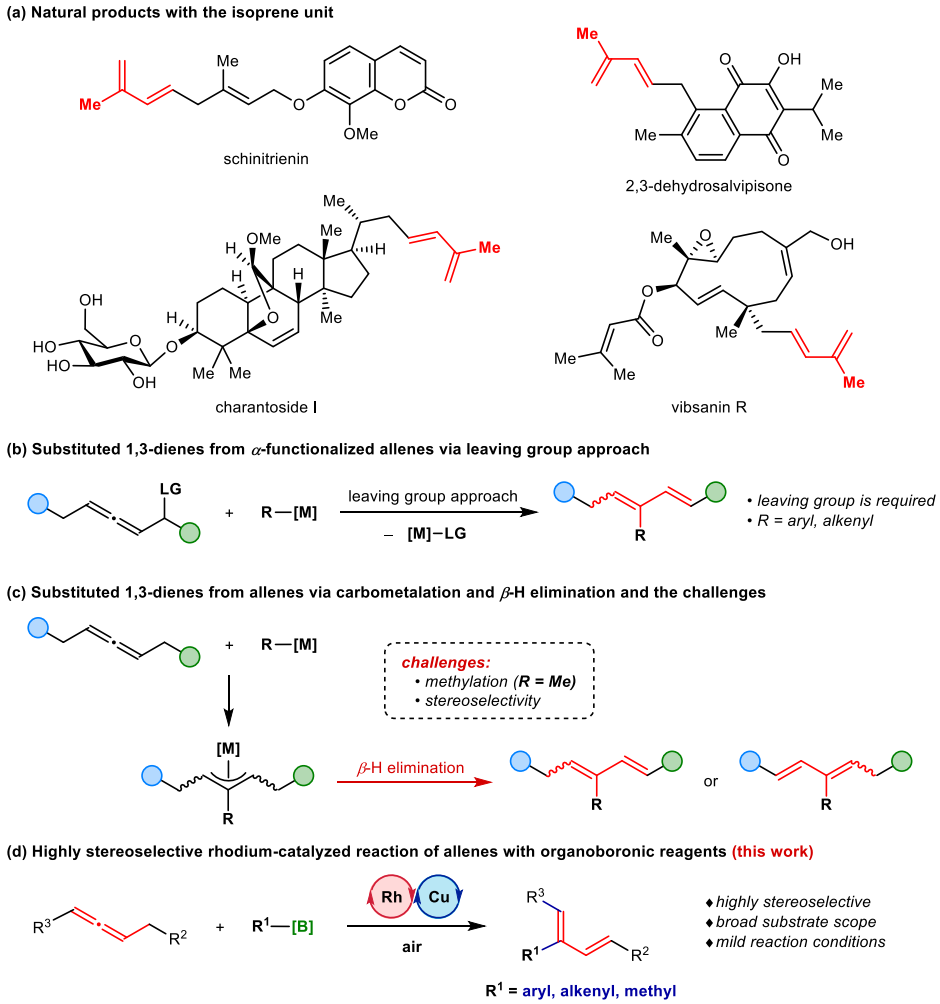


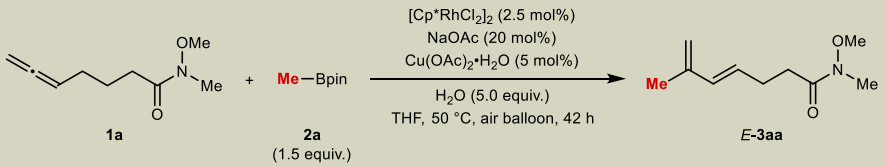
Fig. 1 | Natural products bearing the terminal isoprene unit and synthesis of branched 1,3-dienes from allenes and organometallic species. a Natural products with the terminal isoprene unit. **b** Substituted 1,3-dienes from α-functionalized allenes via a leaving group approach. **c** Substituted 1,3-dienes from allenes via carbometalation and β-H elimination and the challenges. **d** Highly stereoselective rhodium-catalyzed reaction of allenes with organoboronic reagents (this work).

(2.5 mol%), NaOAc (20 mol%), $Cu(OAc)_2{\cdot}H_2O$ (5 mol%), and H_2O (5.0 equiv.) in THF at 50 °C under air atmosphere, *E*-**3aa** was afforded with 57% NMR yield (entry 1). The reactions in other typical solvents were of lower efficiency (entries 2–8). The amount of water has a critical impact on the reactivity (entries 1 and 9–11). Other bases showed lower efficiency as compared to NaOAc (see Supplementary Information, Supplementary Table 1, entries 12–16). The reaction under a pure oxygen atmosphere was in a similar efficiency as compared with the reaction under an air atmosphere (compare entry 12 with entry 1). Under a nitrogen atmosphere (entry 13) or in the absence of $Cu(OAc)_2{\cdot}H_2O$ (entry 14), only 9% or 13% NMR yield of *E*-**3aa** was observed, respectively, indicating that the presence of both air and Cu cocatalyst is critical for the catalytic cycle. To further verify the role of the copper salt, a control experiment was conducted using 2.0 equiv. of $Cu(OAc)_2$ under a nitrogen atmosphere, which also worked and afforded a 45% NMR yield (see Supplementary Information, Reaction development section). Moreover, using 2.0 equiv. of AgOAc or MnO_2 as the oxidant instead of the combination of $Cu(OAc)_2{\cdot}H_2O$ and air, the catalytic transformation could also proceed albeit in lower efficiency, further indicating an aerobic regeneration of the Rh catalyst (see Supplementary Information, Reaction development section). In the absence of catalyst $[Cp^*RhCl_2]_2$, the reaction did not occur (entry 15). NaOAc also plays a vital role (entry 16). The efficiency dropped when the reaction was conducted at room temperature (entry 17; for details, see Supplementary Information, Supplementary Table 1). Other methyl boronic reagents, such as $MeB(OH)_2$ (entry 18) and $MeBF_3K$ (entry 19), gave lower yields. Further optimization led to the observation that with catalyst $Cp^*Rh(OAc)_2$ replacing the combination of $[Cp^*RhCl_2]_2$ and NaOAc, the efficiency was significantly improved with a 70% NMR yield of *E*-**3aa** (entry 20). After further evaluation of various reaction parameters using catalyst $Cp^*Rh(OAc)_2$, we found the amount of water also has a critical impact on the reactivity, and the standard conditions for the methylation have been defined as follows: In the presence of $Cp^*Rh(OAc)_2$ (5 mol%), $Cu(OAc)_2{\cdot}H_2O$ (5 mol%), and H_2O (10.0 equiv.), the reaction of allene **1a** with methyl pinacol boronic ester **2a** in THF at 50 °C under an air atmosphere afforded *E*-**3aa** in 75% NMR yield (entry 21, defined as Method A). (More results and details of the reaction optimization are given in Supplementary Information, Reaction Development section.)

The scope of the methylation was then explored (Fig. 2a). Allenes bearing various synthetically versatile functional groups, such as Weinreb amide (*E*-**3aa,** *E*-**3ay,** *E*-**3az**), protected amines (*E*-**3la**), hydroxy (*E*-**3ca,** *E*-**3ga,** *E*-**3cy**), bromo (**3ta**), could undergo this transformation smoothly, forming an iconic isoprene structural unit. Furthermore, 1,1-disubstituted allene **3va** and 1,3-disubstituted allenes **3wa, 3pa**, and **3ta** all worked well in high yields. Trisubstituted allene **1x** and tetrasubstituted allene **1y** did not work in this reaction with 83% and 98% recoveries, respectively. Interestingly, besides MeBpin,

https://doi.org/10.1038/s41467-024-52209-5

Table 1 | Optimization of the reaction conditions and control experiments



entry	deviation from the above conditions	NMR yield of *E*-3aa (%)	recovery of 1a (%)
1	none	57	0
2	1,4-dioxane instead of THF	30	9
3	DME instead of THF	53	4
4	toluene instead of THF	11	18
5	DCE instead of THF	8	32
6	MeCN instead of THF	2	41
7	MeOH instead of THF	16	0
8	DMSO instead of THF	1	76
9	0 equiv. instead of 5.0 equiv. of H_2O	10	20
10	2.0 equiv. instead of 5.0 equiv. of H_2O	24	14
11	10.0 equiv. instead of 5.0 equiv. of H_2O	49	4
12	O_2 balloon instead of air balloon	58	0
13	N_2 instead of air balloon	9	40
14	without $Cu(OAc)_2{\cdot}H_2O$	13	49
15	without $[Cp^*RhCl_2]_2$	0	96
16	without NaOAc	2	75
17	rt instead of 50 °C	4	80
18	$MeB(OH)_2$ instead of MeBpin	15	0
19	$MeBF_3K$ instead of MeBpin	27	0
20	$Cp^*Rh(OAc)_2$ (5 mol%) instead of $[Cp^*RhCl_2]_2$ (2.5 mol%); without NaOAc	70	0
21	$Cp^*Rh(OAc)_2$ (5 mol%) instead of $[Cp^*RhCl_2]_2$ (2.5 mol%); without NaOAc; 10.0 equiv. instead of 5.0 equiv. of H_2O	75 (73)[a]	0
22	$Cp^*Rh(OAc)_2$ (5 mol%) instead of $[Cp^*RhCl_2]_2$ (2.5 mol%); without NaOAc; 20.0 equiv. instead of 5.0 equiv. of H_2O	71	0

Reaction conditions: **1a** (0.2 mmol, 1.0 equiv.), **2a** (0.3 mmol, 1.5 equiv.), $[Cp^*RhCl_2]_2$ (2.5 mol%), NaOAc (20 mol%), $Cu(OAc)_2{\cdot}H_2O$ (5 mol%), and H_2O (5.0 equiv.) in THF (1 mL, 0.2 M) at 50 °C with an air balloon for 42 h. The NMR yields and the recoveries were determined by ^1H NMR analysis of the crude products using mesitylene as the internal standard. [a] Isolated yield of the reaction on 1 mmol scale for 34 h, as shown in the parentheses.

cyclopropyl pinacol boronic ester also performed well in this transformation (*E*-**3ay**, *E*-**3cy**), and ethyl pinacol boronic ester could also give target product *E*-**3az** albeit in a diminished efficiency.

The remarkable generality of this methylation encouraged us to further illustrate the synthetic potential of this reaction by the synthesis of natural product schinitrienin[4] **14** bearing a terminal isoprene unit (Fig. 2b), which was isolated from the root bark of *Zanthoxylum schinifolium*. From (*E*)-3-methylhept-2-en-6-yn-1-ol, applying bromination followed by etherification with 7-hydroxy-8-methoxycoumarin **11** afforded terminal alkyne **12**. The structure of **12** was further confirmed by X-ray crystallographic analysis. By executing ATA (allenation of terminal alkynes)[41,42], allene **13** was synthesized from terminal alkyne **12**. Finally, the methylation of allene **13** was successfully achieved by the current method, affording schinitrienin **14** concisely. (For more details, see Supplementary Information.)

To our delight, this protocol may also be readily extended to the synthesis of 2-aryl or -alkenyl substituted 1,3-dienes by using aryl or alkenyl boronic reagents (Fig. 3). The reactivity is much higher since the reaction using $[Cp^*RhCl_2]_2$ with NaOAc occurs at room temperature and gives higher yields as compared with that using $Cp^*Rh(OAc)_2$ without NaOAc (see *E*-**3db**). Such branched 1,3-dienes are also versatile and ideal platform for organic synthesis[43–60]. Firstly, we conducted the reactions of monosubstituted allenes with various organoboronic reagents (Fig. 3a), affording 2-substituted 1,3-dienes. Both pinacol boronic esters (Method B) and boronic acids (Method C) could be applied and the generality of pinacol boronic esters is better. Both electron-rich and electron-deficient aryl pinacol boronic esters reacted with allenes in high yields. Importantly, a variety of synthetically versatile functional groups including hydroxy (*E*-**3cb**, *E*-**3gb**, *E*-**3co**, *E*-**3cp**, *E*-**3fq**, *E*-**3cs**), protected hydroxy (*E*-**3hb**), protected amines (*E*-**3kb**, *E*-**3lb**), ester groups (*E*-**3dd**, *E*-**3nl**, *E*-**3nm**, *E*-**3nn**), nitro (*E*-**3de**), methoxy (*E*-**3mf**, *E*-**3af**), Weinreb amides (*E*-**3af**, *E*-**3ac**, *E*-**3ag**, *E*-**3ah**, *E*-**3ai**, *E*-**3aj**, *E*-**3ak**, *E*-**3aab**, *E*-**3aac**, *E*-**3at**, *E*-**3au**), bromo (*E*-**3ag**, *E*-**3ah**), chloro (*E*-**3ai**, *E*-**3aj**), fluoro (*E*-**3ak**), cyano (*E*-**3nm**), trifluoromethyl (*E*-**3nn**), and formyl (*E*-**3cp**) were all compatible. Biphenyl (*E*-**3co**) and naphthyl (*E*-**3fr**) pinacol boronic esters all performed well affording the products in high yields. Heteroaryl on allenes (*E*-**3ib**, *E*-**3bb**, *E*-**3mf**), or heteroaryl pinacol boronic esters (*E*-**3daa**, *E*-**3aab**, *E*-**3aac**), were all compatible for this transformation. The reaction could be easily conducted on a gram scale (*E*-**3ag**). Furthermore, 1-alkenyl pinacol boronic esters could also undergo this transformation to afford substituted [3] dendralenes *E*-**3cs**, *E*-**3at**, and *E*-**3au**, which provides a straightforward, general, and efficient synthetic approach for [3]dendralene synthesis[61,62].

This reaction has also been successfully applied for the modification of complex bioactive molecules (Fig. 3b). Allene **1o**, derived

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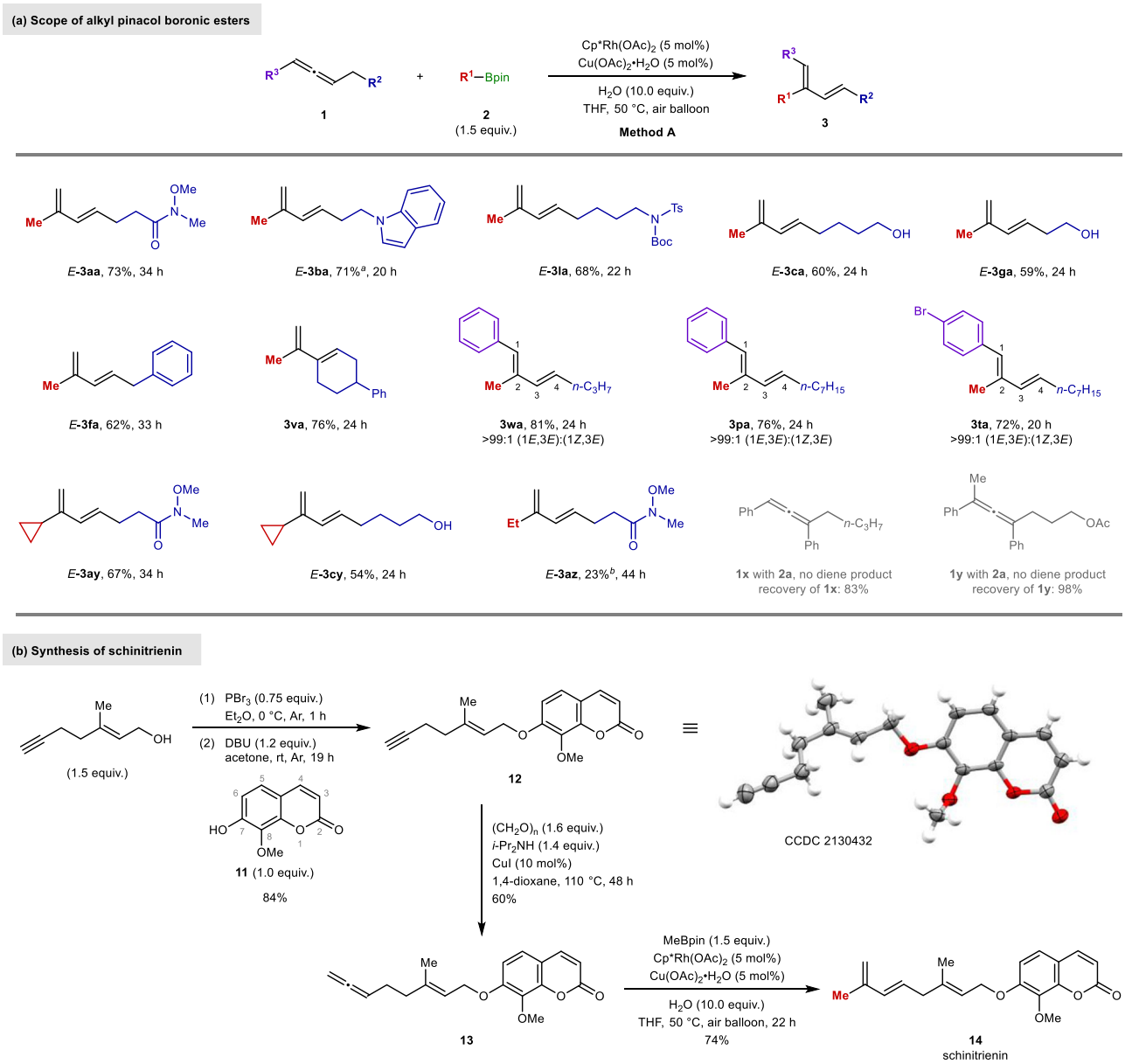


Fig. 2 | Methylation for the construction of the isoprene unit and total synthesis of schinitrienin. a Scope of alkyl pinacol boronic esters. The reactions were conducted with **1** (1.0 mmol, 1.0 equiv.), **2** (1.5 mmol, 1.5 equiv.), Cp*Rh(OAc)$_2$ (5 mol %), Cu(OAc)$_2$•H$_2$O (5 mol%), and H$_2$O (10.0 equiv.) in THF (5 mL, 0.2 M) at 50 °C with an air balloon. The yields were isolated yields. The ratios of (1*E*,3*E*):(1*Z*,3*E*) were determined by ^{1}H NMR analysis of the crude products. **b** Synthesis of schinitrienin. [a] The reaction was carried out on 0.5 mmol scale. [b] Cp*Rh(OAc)$_2$ (10 mol%) was used.

from indomethacin, afforded 1,3-diene *E*-**3oi** in 93% yield. Estrone and 18β-glycyrrhetinic acid were incorporated into corresponding pinacol boronic esters, which gave 1,3-dienes *E*-**3nv** and *E*-**3fw** in 78% and 80% yields, respectively. Furthermore, we examined the substrate scope of 1,3-disubstituted allenes (Fig. 3c). Interestingly, the trisubstituted carbon-carbon double bond in the products was in an excellent *Z*-selectivity, which has been established by X-ray crystallographic analysis of **3tf**. The electronic property of R^1 (**3pb, 3pf** and **3px**) or R^3 (**3rf, 3sf** and **3tf**) did not have an obvious impact on the stereoselectivity[34,38]. Moreover, 1,1-disubstituted allene **1 v** could also undergo this transformation, affording **3vb** in 68% yield.

The synthetic potentials of the product *E*-**3ag** have been illustrated via the [4 + 2] cycloaddition reactions as shown in the Fig. 4. The cycloaddition of *E*-**3ag** with phenylhydroxylamine[63] afforded the heterocyclic compound **15** in 89% yield with an exclusive regioselectivity. The Diels-Alder reactions with dimethyl acetylenedicarboxylate[64], *N*-methylmaleimide, and maleic anhydride gave **16** in 79% yield, **17** in 91% yield, and **18** in 75% yield, respectively, with an excellent diastereoselectivity. The relative configurations of **17** and **18** have all been established by X-ray crystallographic analysis.

To unveil the mechanism, a series of deuterium-labeling experiments were conducted (Fig. 5a–c). The reaction of 4,4-di-deuterated monosubstituted allene **1d**-d_2 with phenylboronic acid **2b'** under the standard conditions afforded *E*-**3db**-d_3-1, in which the deuterium at the C(4)-position lost partially (84% deuterium incorporation), and the C(1)-position was partially deuterated (53% and 51% deuterium incorporation) (Fig. 5a). When 1,1-di-deuterated allene **1d**-d_2' was reacted with **2b'** under the standard conditions, *E*-**3db**-d_3-2 was afforded with 81% and 81% deuterium incorporation on C(1)-position, and 20% deuterium incorporation on C(4)-position (Fig. 5b). It is noted that there should be a large amount of active proton source (water, boronic acid, etc.) in the ambient environment, which did not seem to interfere with

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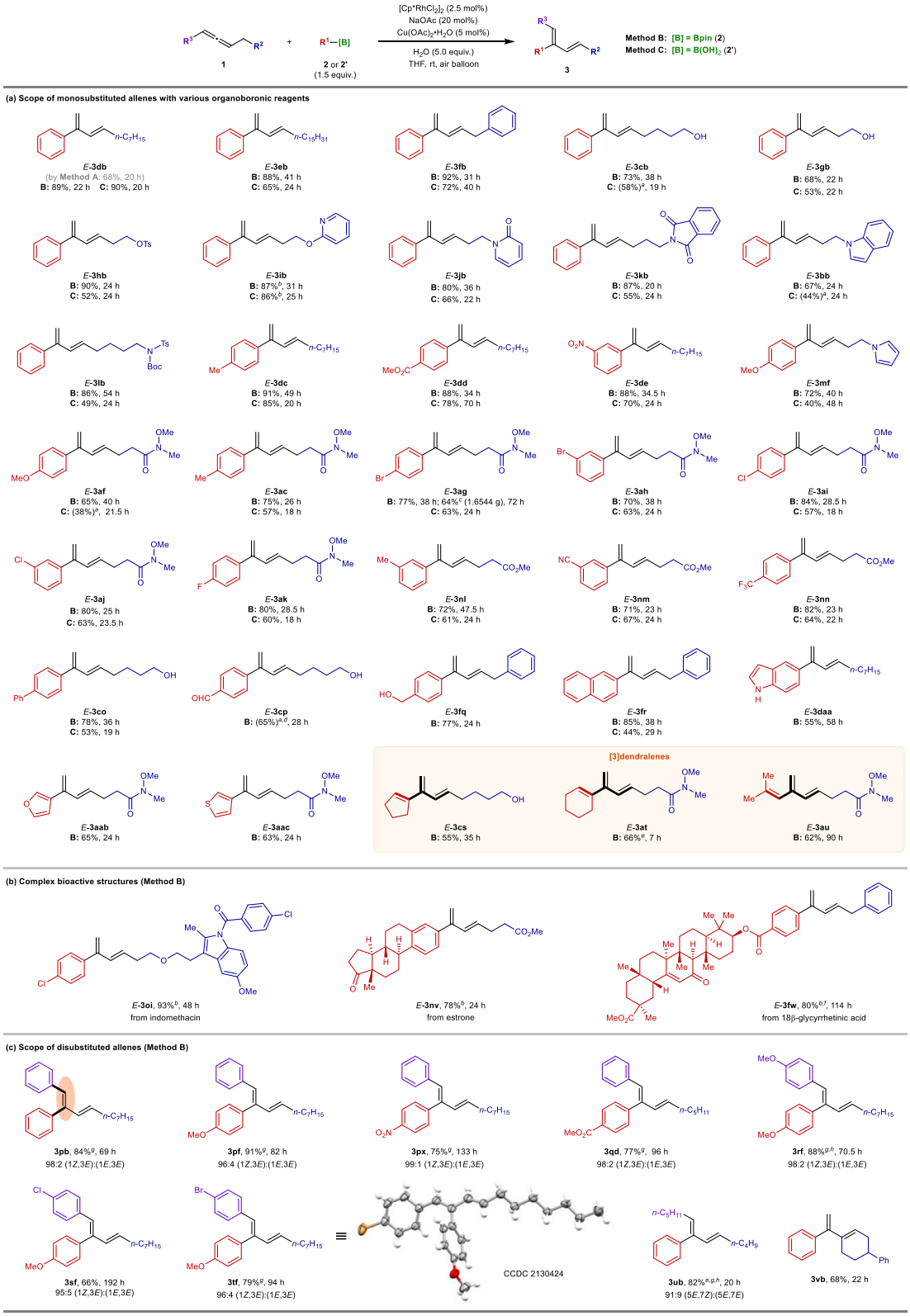


Fig. 3 | Substrate scope of the reaction of allenes 1 with aryl or alkenyl boronic reagents 2 or 2'. The reactions were conducted with **1** (1.0 mmol, 1.0 equiv.), **2** or **2'** (1.5 mmol, 1.5 equiv.), $[Cp^*RhCl_2]_2$ (2.5 mol%), NaOAc (20 mol%), $Cu(OAc)_2{\cdot}H_2O$ (5 mol%), and H_2O (5.0 equiv.) in THF (5 mL, 0.2 M) at room temperature with an air balloon. The yields were isolated yields. The ratios of (*Z*,*E*):(*E*,*E*) were determined by ^{1}H NMR analysis of the crude products. **a** Scope of monosubstituted allenes with various organoboronic reagents. **b** Complex bioactive structures (Method B). **c** Scope of disubstituted allenes (Method B). [a] NMR yield, as shown in the parentheses, which was determined by ^{1}H NMR analysis of the crude product using dibromomethane as the internal standard. [b] The reaction was carried out on 0.2 mmol scale. [c] The reaction was carried out on 8 mmol scale. [d] The aldehyde was unstable during the isolation, we characterized the corresponding alcohol after reduction with $NaBH_4$. [e] The reaction was carried out at 50 °C. [f] THF (0.1 M), and there was 6% recovery of **1f**, which was determined by ^{1}H NMR analysis of the crude product using dibromomethane as the internal standard. [g] $[Cp^*RhCl_2]_2$ (5 mol%) was used. [h] The reaction was carried out on 0.5 mmol scale.

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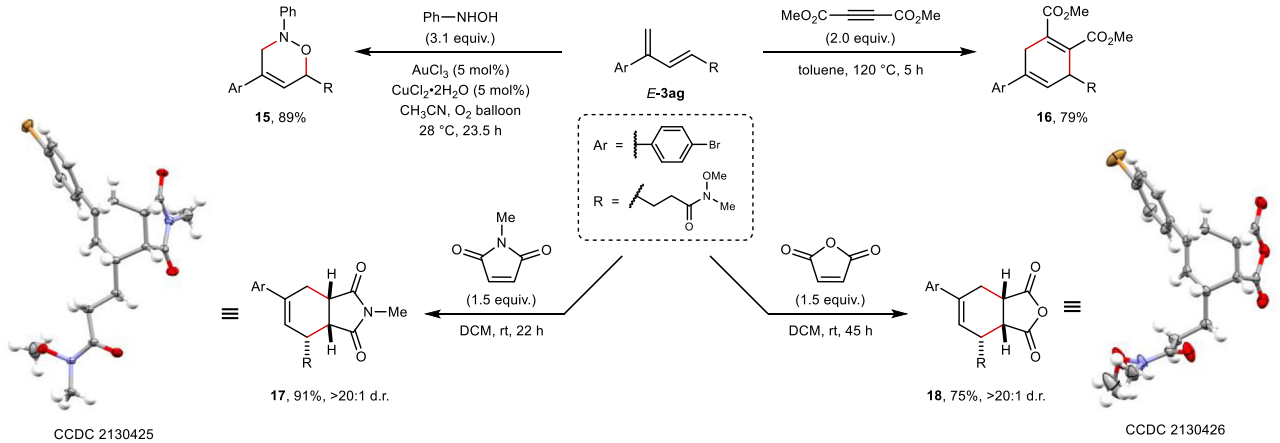


Fig. 4 | Synthetic potentials of the product *E*-3ag. [4 + 2] Cycloaddition reactions of *E*-**3ag** with phenylhydroxylamine, dimethyl acetylenedicarboxylate, *N*-methylmaleimide, and maleic anhydride, respectively.

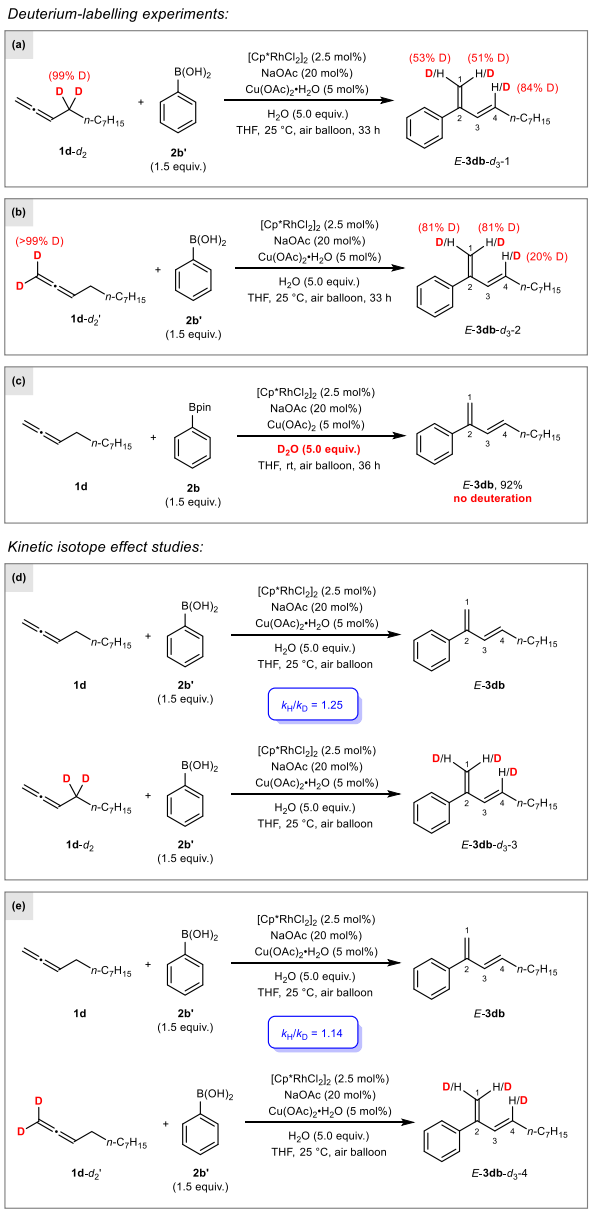


Fig. 5 | Mechanistic studies. a–c Deuterium-labeling experiments. **d–e** Kinetic isotope effect studies.

the [Rh]–D/H species since replacement of H_2O and $Cu(OAc)_2$•H_2O with D_2O and $Cu(OAc)_2$ under the standard conditions for the reaction of non-deuterated allene **1d** with phenyl pinacol boronic ester **2b** afforded non-deuterated product *E*-**3db** (Fig. 5c). The kinetic isotope effect (KIE) studies using **1d**-d_2 (Fig. 5d) and **1d**-d_2′ (Fig. 5e) demonstrated k_H/k_D values of 1.25 and 1.14, respectively. These observed low KIE values suggest that the β-H elimination is not the turnover-limiting step. The different KIE values indicate a minor difference upon involving the D atom at 1 or 4 position in **1d**-d_2 or **1d**-d_2′.

Based on these results, a catalytic cycle is proposed (Fig. 6). At first, the $[Cp^*RhOAc]^+$ species **A** undergoes transmetalation with the organoboronic reagent to generate $[Cp^*RhR^1]^+$ species **B**, which would undergo subsequent highly stereoselective carbometalation with allene **1** to generate the stereodefined η^3-allylic rhodium intermediate *anti*-**C**. Its β-H elimination would afford the complex of the diene and $[Cp^*RhH]^+$ species, intermediate **D**. The $[Cp^*RhH]^+$ species in intermediate **D** can undergo hydrometalation with the 1,3-diene, generating π-allylic intermediate **E** or *anti*-**C**, delivering the hydride to the C(1)- or C(4)-position of the 1,3-diene unit. Reductive elimination and dissociation of the diene from the complex **D** in the presence of AcO^- would release product **3**, AcOH, and Cp*Rh. Finally, Cp*Rh could be oxidized under the catalysis of Cu with the oxygen in air as the oxidant to regenerate the catalytically active Rh(III) species **A**, finishing the catalytic cycle. In addition, it is interesting that the acetate works in a catalytic way, both AcO^- and AcOH would consume and regenerate during the reaction via the reaction of AcOBpin with water, the reductive elimination, and the catalytic oxidation cycle.

The result of the deuterium-labeling experiment using **1d**-d_2 in Fig. 5a could be further explained as shown in Fig. 7a. The overall product *E*-**3db**-d_3-1 was a mixture of **P1** – **P6**, thus, the deuterium incorporation of C(1)- and C(4)-positions was a combined result of these six different deuterated products **P1** – **P6**: Firstly, the $[Cp^*RhPh]^+$ species and allene **1d**-d_2 undergo carbometalation to generate the η^3-allylic rhodium intermediate η^3-**F1**. The subsequent *syn*-β-D elimination would generate diene-Rh coordination complex **F2**, which could undergo dissociation of the diene and reductive elimination to release monodeuterated product **P1**, AcOD, and Cp*Rh in the presence of AcO^-. The $[Cp^*RhD]^+$ moiety in complex **F2** could also undergo *syn*-deuterometalation with the terminal C = C bond in 1,3-diene generating π-allylic intermediate η^3-**F3**. The *syn*-β-H elimination of η^3-**F3** could generate new diene-Rh coordination complexes **F4** and **F8**, which could release two different 1,4-dideuterated products **P2** and **P3**, AcOH, and Cp*Rh in the presence of AcO^- via dissociation of the diene and reductive elimination. **F4** and **F8** could also undergo further *syn*-hydrometalation to generate allylic metallic species η^3-**F5** and η^3-**F9**,

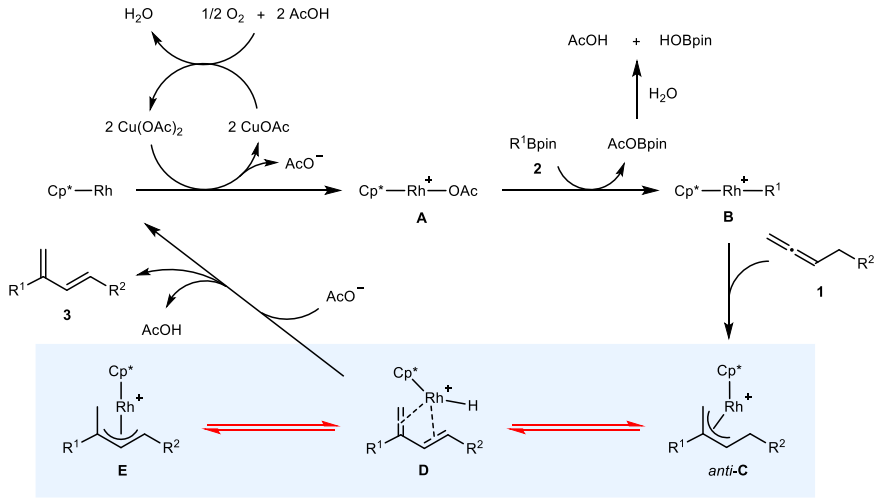


Fig. 6 | Proposed mechanism. Possible reaction mechanism of allenes with pinacol boronic esters.

which could also undergo *π*-*σ*-*π* rearrangement to generate new allylic Rh species η^3-**F6** and η^3-**F10**. Subsequent *syn*-*β*-D elimination would generate new diene-Rh coordination complexes **F7** and **F11**, which could release monodeuterated diene product **P4** and **P5**, AcOD, and Cp*Rh in the presence of AcO^-. The *syn*-deuterometalation of **F7** and **F11** may also generate allylic Rh species η^3-**F12**. The *syn*-*β*-H elimination of η^3-**F12** would generate complex **F13**, which could release terminally dideuterated product **P6**, AcOH, and Cp*Rh in the presence of AcO^-. The loss of deuterium incorporation was calculated as $2 \times 0.99 - 0.53 - 0.51 - 0.84 = 0.10$ D. The loss of deuterium occurred during the formation of **P1, P4**, and **P5** by the release of AcOD to the ambient environment.

The rationale for the result of the deuterium-labeling experiment using **1d**-d_2' in Fig. 5b is similar (see Supplementary Information, Supplementary Fig. 7), and the loss of deuterium incorporation was calculated as $2 - 0.81 - 0.81 - 0.20 = 0.18$ D. The loss of deuterium occurred in the formation of **P2', P3'**, and **P6'** by the release of AcOD to the ambient environment. The different extent of the deuterium scrambling in these two deuterium-labeling experiments may be caused by the observed small kinetic isotope effect in different *syn*-*β*-H/D elimination, *syn*-hydro-/deuterometalation, and reductive elimination of [Cp*Rh-H/D]$^+$ species.

The high stereoselectivity of the reaction of 1,3-disubstituted allenes could be attributed to the steric effect: The *π*-*σ*-*π* rearrangement from η^3-(*syn*,*anti*)-**G1** to η^3-(*anti*,*anti*)-**G2** is disfavored due to the large allylic 1,3-strain as well as the closer distance between Ar^2 and the Rh-Cp* moiety[65,66]. Based on the information shown in Fig. 7a, there should be another type of allylic Rh intermediate η^3-(*syn*,*syn*)-**H1**, which would isomerize to η^3-(*syn*,*syn*)-**H2** via the C–C bond rotation, however, due to the increased allylic 1,3-strain[65,66], such isomerization is also disfavored. Overall, the formation of (1*E*,3*E*)-stereoisomers would be disfavored (Fig. 7b).

To gain a comprehensive understanding of the reaction mechanism, we conducted detailed density functional theory (DFT) calculations at the $SMD_{(THF)}$/M06/6-311+g**,SDD//M06/6-31 g*,LANL2DZ level to investigate the reaction energy profiles, utilizing penta-1,2-diene **1'** and phenyl pinacol boronic ester **2b** as the model substrates (Fig. 8a) (for more detailed information, see Supplementary Information and Supplementary Data 1). The displayed energies are Gibbs free energies incorporating the solvent effect of THF. The transmetalation of [Cp*RhOAc]$^+$ with **2b** would result in the formation of [Cp*RhPh]$^+$. The subsequent coordination of [Cp*RhPh]$^+$ with **1'** provides the intermediate **Int1**, which was selected as the free energy reference. The initial carbometalation step involves the insertion of the coordinated allenic C=C double bond into the Rh–C bond via a four-membered ring transition state. Due to the possibility of the participation of each C=C bond of penta-1,2-diene **1'**, four competing transition states associated with the selective carborhodation step are obtained, which are denoted as **TS1_a, TS1_b, TS1_c** and **TS1_d**, separately (Fig. 8b). **TS1_a**, which leads to the formation of the thermodynamically favorable *π*-allylic Rh^{III} complex **Int2_*anti***, was found to be the most stable. This step overcomes an energy barrier of 6.6 kcal/mol and exhibits exergonicity of 41.0 kcal/mol, indicating the irreversibility of this step. The subsequent 1,4-hydride transfer proceeds through a stepwise *β*-hydride elimination/hydrometalation mechanism via a bypassed intermediate, diene-[Cp*RhH]$^+$ complex **Int3**. Remarkably, the [Cp*RhH]$^+$ species in **Int3** would undergo hydrometalation with the 1,3-diene, proceeding through **TS3** to yield the *π*-allylic Rh^{III} complex **Int4**, as well as traversing **TS2** to regenerate the *π*-allylic Rh^{III} complex **Int2_*anti***, due to the similar free energy barrier observed for **TS2** and **TS3** relative to **Int3** (2.2 kcal/mol versus 1.9 kcal/mol). Meanwhile, **Int4** could also reproduce **Int3** via **TS3**, by overcoming an energy barrier of 15.3 kcal/mol. Moreover, in the presence of AcO^-, the reductive elimination of **Int5** proceeds through **TS4** by surpassing a free energy barrier of 4.8 kcal/mol, to produce the thermodynamically more stable complex **Int6**. The release of AcOH of **Int6** generates the diene-Cp*Rh^I species **Int7**. Finally, the oxidation of Rh^I species with Cu^{II} in the air regenerates the catalytically active Rh^{III} species and releases the 1,3-diene to complete the catalytic cycle.

In summary, a Rh-catalyzed reaction of allenes with organoboronic reagents has been developed, affording diversified branched 1,3-dienes in an excellent stereoselectivity under very mild reaction conditions, enjoying a very broad substrate scope and tolerating a large variety of functional groups. The reaction uses $Cu(OAc)_2 \cdot H_2O$ as a cocatalyst with the assistance of oxygen from air as the green oxidant to complete the catalytic cycle. Methylation has been achieved by this protocol and the synthetic practicability has been illustrated by the concise synthesis of natural product schinitrienin. A mechanism has been proposed involving reversible *β*-H elimination as well as hydrometalation and isomerization of the allylic rhodium species. Further studies are currently ongoing in our laboratory.

Methods

General procedure for the synthesis of 2-methyl substituted 1,3-dienes 3 (Method A)

To a Schlenk flask were added Cp*$Rh(OAc)_2$ (17.8 mg, 0.05 mmol, 5 mol %), $Cu(OAc)_2 \cdot H_2O$ (10.0 mg, 0.05 mmol, 5 mol%), methyl pinacol

https://doi.org/10.1038/s41467-024-52209-5

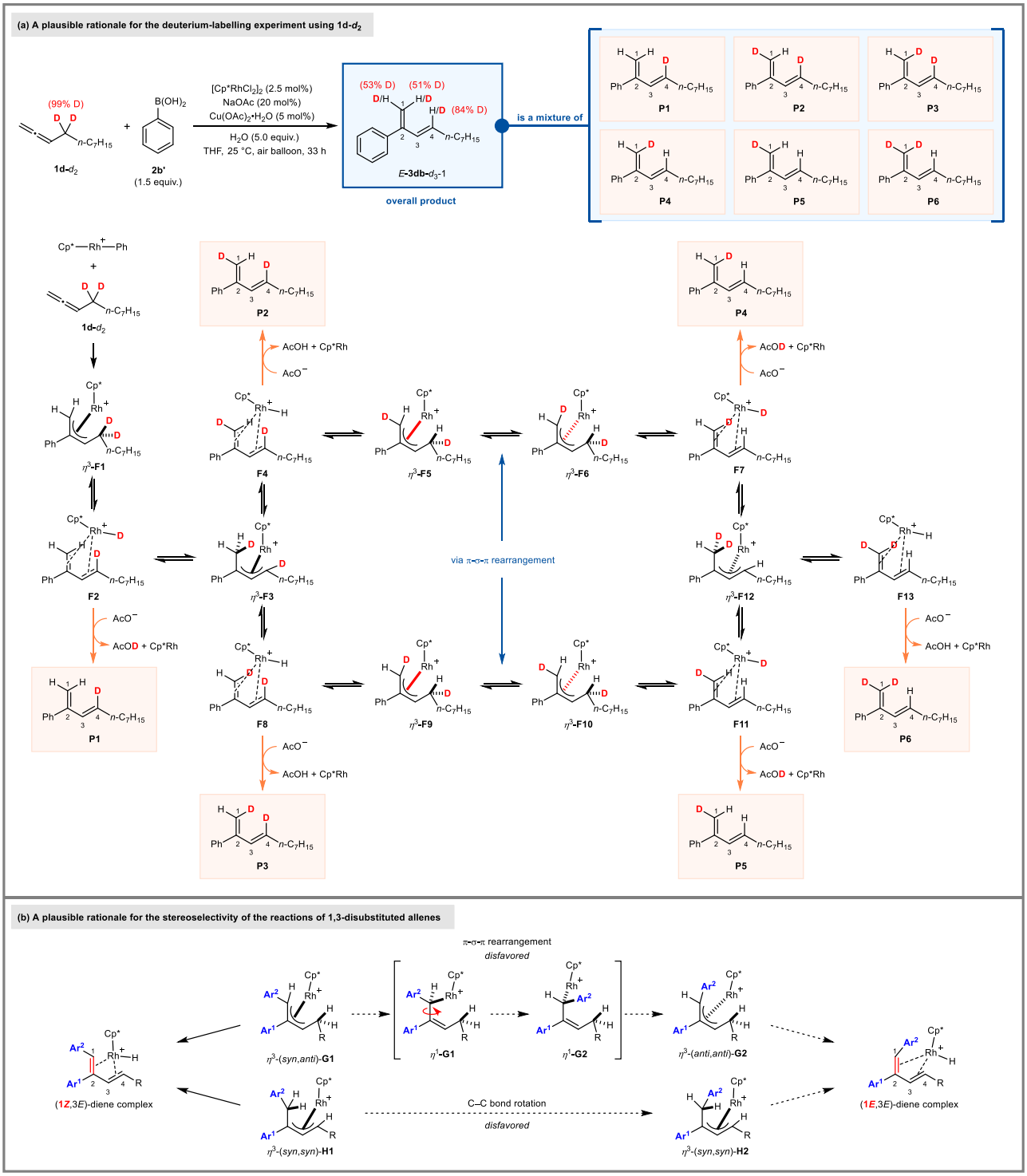


Fig. 7 | Plausible rationales for the deuterium-labeling experiments and the stereoselectivity. a A plausible rationale for the deuterium-labeling experiment using **1d**-d_2. **b** A plausible rationale for the stereoselectivity of the reactions of 1,3-disubstituted allenes.

boronic ester **2a** (213.0 mg, 1.5 mmol, 1.5 equiv.), H_2O (180 µL, 10.0 mmol, 10.0 equiv.), allene **1** (1.0 mmol, 1 equiv.), and THF (5 mL) sequentially without protection from an inert atmosphere. Then the Schlenk flask was connected with an air balloon, and the resulting mixture was stirred at 50 °C until completion of the reaction. The crude reaction mixture was filtered through a short column of silica gel (3 cm), eluted with ethyl acetate (30 mL), and concentrated. The residue was purified by column chromatography on silica gel to afford the pure product **3**.

General procedure for the synthesis of 2-aryl or -alkenyl substituted 1,3-dienes 3 (Methods B and C)

To a Schlenk flask were added $[Cp^*RhCl_2]_2$ (15.5 mg, 0.025 mmol, 2.5 mol%), NaOAc (16.4 mg, 0.20 mmol, 20 mol%), $Cu(OAc)_2•H_2O$ (10.0 mg, 0.05 mmol, 5 mol%), pinacol boronic ester **2** or organoboronic acid **2'** (1.5 mmol, 1.5 equiv.), H_2O (90 µL, 5.0 mmol, 5.0 equiv.), allene **1** (1.0 mmol, 1 equiv.), and THF (5 mL) sequentially without protection from an inert atmosphere. Then the Schlenk flask was connected with an air balloon, and the mixture was stirred at room

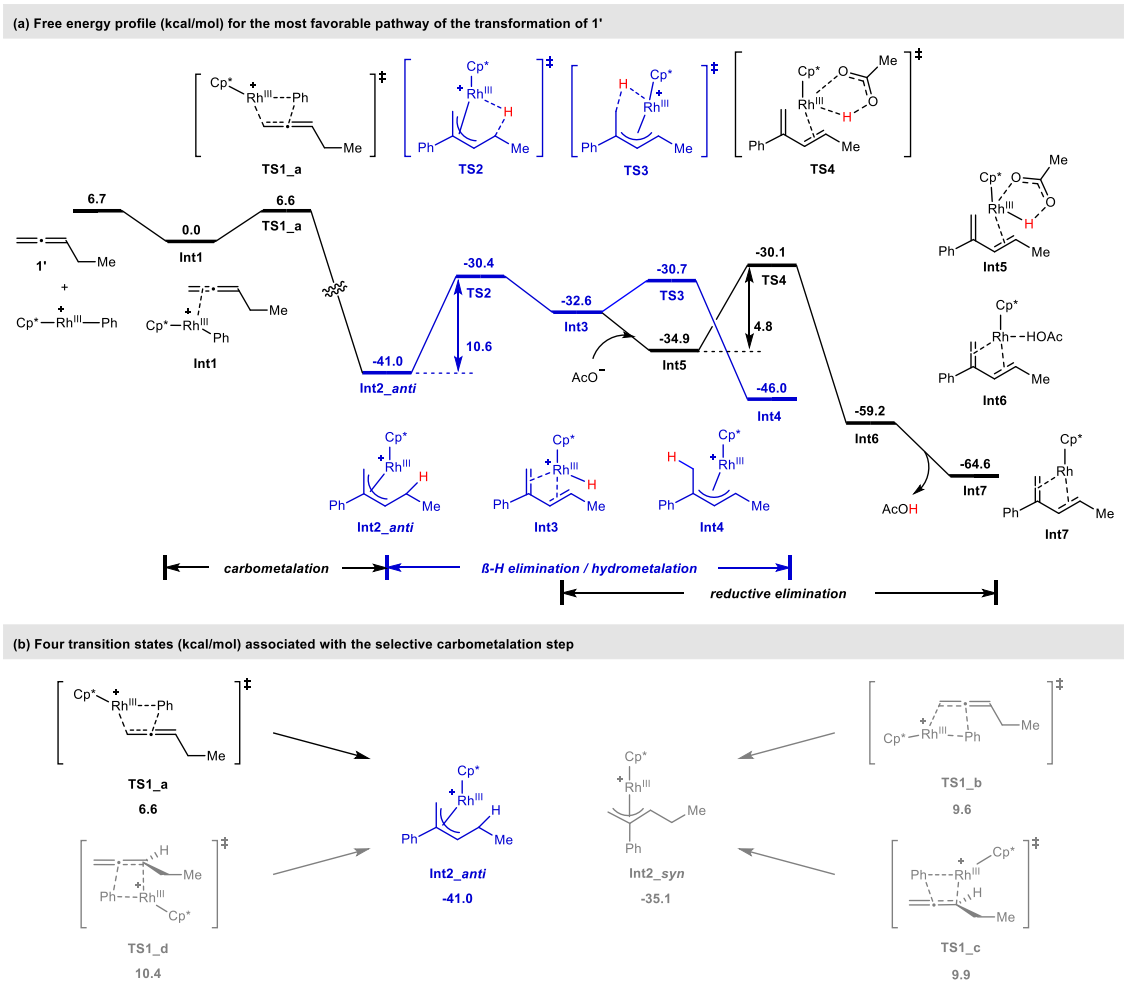


Fig. 8 | DFT calculations. a Free energy profile (kcal/mol) for the most favorable pathway of the transformation of **1'**. **b** Four transition states (kcal/mol) associated with the selective carbometalation step.

temperature until completion of the reaction. The crude reaction mixture was filtered through a short column of silica gel (3 cm), eluted with ethyl acetate (30 mL), and concentrated. The residue was purified by column chromatography on silica gel to afford the pure product **3**.

Data availability

Experimental procedures, characterization of the compounds, and density functional theory calculations are available in the Supplementary Information. Crystallographic data have been deposited at the Cambridge Crystallographic Data Center, under deposition numbers CCDC 2130429 (**6**, in Supplementary Information), CCDC 2130432 (**12**), CCDC 2130424 (**3tf**), CCDC 2130425 (**17**) and CCDC 2130426 (**18**). Copies of the data can be obtained free of charge via https://www.ccdc.cam.ac.uk/structures/. Data supporting the findings of this manuscript are also available from the authors upon request.

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Acknowledgements

Financial support from National Key R&D Program of China (Grant No. 2022YFA1503200 for S.M.), the National Natural Science Foundation of China (Grant No. 21988101 for S.M. and Grant No. 21801041 for H.Q.), and Shanghai Rising-Star Program (23QA1400400 for H.Q.) is greatly appreciated. We thank Mr. Zhengnan Zhou in this group for reproducing the results of *E*-**3gb** (method B), *E*-**3at**, and **3pb** presented in Fig. 3.

Author contributions

S.M. and H.Q. directed the research. S.M., H.Q., and W.W. designed the experiments. W.W. and W.H. performed the experiments. S.H. and X.Z. performed the density functional theory calculations. W.W., S.H., X.Z., H.Q., and S.M. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-024-52209-5.

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Peer review information *Nature Communications* thanks Raquel Soengas, and the other, anonymous, reviewers for their contribution to the peer review of this work. A peer review file is available.

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